

CNS2025: Homework 14

Due: 2025-12-10 23:59

Consider the rectification nonlinearity model described in [Slide 10](#) of the lecture:

$$\tau_r \frac{dv(\theta)}{dt} = -v(\theta) + \left[h(\theta) + \frac{\lambda_1}{\pi} \int_{-\pi}^{\pi} d\theta' \cos(\theta - \theta') v(\theta') \right]_+$$

with $\lambda_1 = 1.9$.

Exercise 1

Use the following code to generate the input $h(\theta)$:

```
dth = 0.1 # Angle resolution
ths = np.arange(-np.pi, np.pi, dth) # Angles
h0 = np.cos(ths + np.pi/2) - 0.8
h0[h0 < 0] = 0
h1 = np.cos(ths - np.pi/2) - 0.82
h1[h1 < 0] = 0
h = h0 + h1
k = np.exp(-np.linspace(-10, 10, int(2//dth))**2/10) # Smoothing filter
k /= np.sum(k) # Normalize
h = np.convolve(h, k, 'same')
```

Produce the plots as shown in [Slide 20](#) to verify the model behaves as a winner-takes-all processor.

Exercise 2

Write your own Python code to generate three inputs to approximate the three with different offset levels as shown in Plot A of [Slide 21](#). Reproduce the the Plot B to verify the output is gain modulated by the offset level of the input.